Q.24 Explain the advantages of using a PLC over Relays.

Q.25 What are the different types of PLC registers?

Q.26 Write the applications of micro controllers.

Q.27 How a micro controller is different from a micro processor?

Q.28 Explain the role and types of input and Output devices used with PLCs.

Q.29 Describe the various memory used in a micro controller.

Q.30 What are the different types of addressing modes in a micro controller?

Q.31 Differentiate between Assembler and compiler.

Q.32 Discuss the advantages of a micro controller.

Q.33 Write a short note on serial communication in a micro controllers.

Q.34 Explain any two arithmetic instructions of PLC.

Q.35 Explain the use of ladder logic programming with an example.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the architecture of an 8051 micro controller in details.

Q.37 Explain the pin diagram of a 8051 microcontroller with neat diagram.

Q.38 Describe the various building blocks of PLC and draw their diagrams also.

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Roll No. ....

5th Sem / Elect.

Sub : Programmable Logic Controllers & Micro Controllers

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is an example of a PLC output device?

a) Sensor b) Motor

c) Push Button d) Switch

Q.2 Which type of PLC is preferred for small scale applications?

a) Modular PLC b) Compact PLC

c) Distributed PLC d) Redundant PLC

Q.3 What is the role of an ADC in a micro controller?

a) Convert digital signals to analog signals

b) Convert analog signals to digital signals

c) Store digital signals

d) Process arithmetic calculations

Q.4 In a micro controller, an interrupt is used to :

a) Delay execution of the program

b) Stop the execution permanently

c) Execute a task immediately when an event occurs

d) Increase the clock speed

Q.5 Which of the following is the primary advantage of using a PLC in industrial automation?
a) Low power consumption
b) High speed internet connectivity
c) Ability to handle complex automation tasks with reliability
d) Large data storage capability

Q.6 In a PLC, what does a counter do?
a) Measures the voltage of the circuit
b) Counts the number of events occurring in a system
c) Controls the speed of a motor
d) Stores the program in the memory

Q.7 Which of the following is the main advantage of using a micro controller?
a) High storage capacity
b) Low power consumption and compact size
c) Ability to run multiple operating systems
d) High speed processing like a super computer

Q.8 Which component in a micro controller is responsible for performing arithmetic and logic operations?
a) RAM b) ALU
c) Control Unit d) GPIO Ports

Q.9 Which bus is used to transfer data between the micro controller and external memory?
a) Control bus b) Address bus
c) Data bus d) Power bus

Q.10 Which of the following communication protocols is NOT used in micro controllers?
a) UART b) SPI
c) TCP/IP d) I2C

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The register that stores the address of the next instruction to be executed is called the ______.

Q.12 The main component of a micro controller is the hard disk (T/F)

Q.13 Microcontrollers consume less power compared to micro processors (T/F)

Q.14 The ALU in a micro controller only performs multiplication operations (T/F)

Q.15 The clock speed of a micro controller determines the number of _____ executed per second.

Q.16 Interrupts help a micro controller to respond _____ to external events.

Q.17 Which programming language is most commonly used in PLCs?

Q.18 What is the role of the CPU in a PLC?

Q.19 What is the function of an ADC in a micro controller?

Q.20 Which bus is used to transfer data in a micro controller?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Discuss the applications of PLCs in industries.

Q.22 Explain the working of timers and counters in PLCs.

Q.23 Describe the role of SCADA in industry.